

INTRODUCTION TO PYTHON

Welcome to this module on the python development environment. This module will cover what python programming language is, the components of python development environment, setting up python development environment and creating your first program.

What is python

Python is a free high level programming language with very nice syntax. High level languages are close to human language and they are independent from the various platforms such as Windows, Linux and Mac OS.

Python comes with many libraries that can be used to create a range of applications. Python is also available in many platforms such as windows, linux and Mac OS.

Python is one of the common programming languages used in organizations today.



The image is copyrighted

Where python is being used

1. Academia: Python is used in many academic institutions to introduce learners to programming
2. Python has scientific tools that can be used for linear algebra and optimization and integration
3. Machine learning: Python has many libraries for carrying out machine learning tasks
4. Statistics
5. Data analysis

Program

A program is a set of instructions that tells the computer what to do. Programs are used in various ways such as supermarkets, banks, hospitals, schools and many others.

Programming

The act of writing the set of instructions to computers is called programming. Some people refer to it as coding.

Programming language

Programming languages are the means with which humans write instructions to computers

Machine language

Is a low level language that is understandable by computers. All other languages need to be translated to machine language.

Development environment components

Compiler

Converts the whole program written in a high level language to machine language. Compilers link the codes analysed

Interpreter

Interpreters convert single lines to machine language. Interpreters do not link the codes.

Editor

These are softwares or programs that enable programmers to write and edit

IDE (Integrated development environment):

A development environment is where the programmer writes the instructions. An IDE contains both an editor and a compiler. An editor is a program like a text editor where you can write your code. A compiler is a program that translates the instructions you have written to instructions that computers understand. Computers understand only 1's and 0's, therefore there is a need to convert the source code(Instructions you have written in a high level language) to machine code (Code that machines understand).

Components of a python program

Statements:

Instructions that python interpreter can execute are called statements

Comments:

Comments are instructions or explanation that a programmer provides for the purpose of making their code readable and easy to follow. Comments are ignored by the python interpreter.

Comments in python are preceded by the # symbol

To write multiple lines comments we enclose the lines in `"""` quotes

Indentation

Python organizes code with indentation, you should be very keen with how you use indentation. More about indentation will be handled when we talk about loops, conditions and branching.